

CSC 296 Assignment

1. A project has the following list of activities, their predecessors, and durations in days:

Activity	Predecessor	Duration
A	-	3
B	A	4
C	A	2
D	B	5
E	C	1
F	C	2
G	D, E	4
H	F, G	4

Instructions:

- Draw the project network diagram showing all activities and dependencies.
- Perform forward and backward pass calculations to determine:
 - Earliest Start (ES), Earliest Finish (EF)
 - Latest Start (LS), Latest Finish (LF)
- Determine the critical path and state the total project duration.
- Identify any activity that has slack time, and calculate the amount of slack.

2. A small-scale entrepreneur produces two types of fruit drinks: Mango Juice and Orange Juice.

- Each litre of Mango Juice yields a profit of ₦500.
- Each litre of Orange Juice yields a profit of ₦300.
- Each litre of the drinks requires the following resources:

Resource	Mango juice	Orange juice	Maximum available
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Sugar(kg)	2	1	40kg
Water(litres)	3	4	60litres

Let: x = litres of Mango Juice produced:

y = litres of Orange Juice produced

Instructions:

- Formulate the objective function and the constraints.
- State the optimal solution and the maximum profit.